Hyperspectral Image Compression via Cross-Channel Contrastive Learning

Yuanyuan Guo, Yanwen Chong, and Shaoming Pan

***Abstract*— In recent years, advances in deep learning have greatly promoted the development of hyperspectral image (HSI) compression algorithms. However, most existing compression approaches directly rely on rate–distortion (RD) optimization without other guidance during model learning. Therefore, this brings challenges to distinguishing similar features or objects that are widely available in HSIs, especially in remote sensing scenes, since quantification in lossy compression can cause informative attribute (e.g., category) collapse and loss problems at high compression ratios. In this article, we propose a novel hyperspectral compression network via contrastive learning (HCCNet) to help generate discriminative representations and preserve informative attributes as much as possible. Specifically, we design a contrastive informative feature encoding (CIFE) to extract and organize discriminative attributes from the original HSIs by enlarging the discrimination over the learned latents in different channel indexes to relieve attribute collapses. In the case of attribute losses, we define a contrastive-invariant feature recovery (CIFR) to discover the lost attributes via contrastive feature refinement. Experiments on five different HSI datasets illustrate that the proposed HCCNet can achieve impressive compression performance, such as improvement of the peak signal-to-noise ratio (PSNR) from 28.86 dB [at 0.2284 bit per pixel per band (bpppb)] to 30.30 dB (at 0.1960 bpppb) on the Chikusei dataset.**

***Index Terms*— Contrastive learning, deep learning, high-quality reconstruction, hyperspectral image (HSI) compression.**

I. Introduction

HYPERSPECTRAL images (HSIs) characterize the observed materials by encompassing dozens or even hundreds of spectral bands. Therefore, they enable to provide a distinctive spectral signature for the categorization of material captured by each pixel. For that reason, HSIs have been involved in a great number of practical applications, such as precision agriculture [1], geological exploration, and mineralogy [2], among many others. However, the availability of large amounts of information results in great challenges, such as HSI storage and transmission issues due to limited bandwidth availability and storage capacity in real life. Therefore, HSI compression has become a crucial and indispensable technique, with the primary purpose of preserving the scientific information inherent to the image without incurring losses. Such a goal is an important one, but a challenging one as well, and lossless compression is incapable of supporting a high compression ratio, since it needs to preserve information without the creation of distortion. Lossy compression is one method that sacrifices a small amount of information to compensate for this weakness. In this technique, it would be highly desirable to make a trade-off between the entropy [3] of the given compressed discrete data (rate) and the error introduced by quantization (distortion), so that the downstream applications that use this data could be constructed without too much performance degradation.

The early approaches of hyperspectral compression [4], [5] directly encode HSIs band by band, following the red, green, and blue (RGB) image compression patterns, which is unable to take advantage of interband correlation and results in suboptimal rate–distortion (RD) performance. To fully explore interband correlation, many efforts have extended the original 2-D compression algorithms into corresponding 3-D versions [6], [7], [8]. Moreover, to extend the scope and ease of those 3-D compression algorithms, sparse representation [9] and matrix and tensor decomposition [10], [11] are involved to reduce the computational complexity. Among the various methods of HSI lossy compression, this article focuses on transform coding methods [5], [12], [13]: that are commonly operated in a more readily quantified and controlled domain rather than within the pixel domain, with the goal of achieving the optimal rate–distortion performance. In the case of traditional transform coding, HSIs are often represented into a statistically independent space via an invertible linear transform [e.g., the Karhunen–Loève transform (KLT)], which makes it easier to encode each coefficient independently. However, it seems unlikely to achieve statistical independence via linear transforms, since HSIs are not produced by the composition of linear cascades with independent forms [14]. This suggests that a nonlinear transform may be more appropriate for HSIs.

Recently, deep learning has generated a remarkable amount of success in both computer vision and image processing. Similar to traditional transform coding methods, they also transform images into a compact representation before quantization and entropy coding (two other basic modules in transform coding). In contrast, their transforms are cascades of recurrent neural networks (RNNs) or convolutional

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neural networks (CNNs), which are used to approximate nonlinear functions. In addition to the difference in the form of transforms, all modules in the deep learning-based compression methods can be optimized jointly rather than individually. This optimization way can make compression modules more adaptive to each other and thereby push the performance closer to the optimal solution. The pioneering works [15], [16] directly modeled images via a binarized latent representation with higher capacity transforms (e.g., RNNs), so that the separate entropy coding step was naturally removed. However, the binary representation reveals a highly inflexible entropy model (without trainable parameters) that cannot be easily controlled; therefore, it is not clear whether the rate–distortion performance is optimal already. Ballé et al. [17] compensated for that issue by modeling a trainable entropy model with a fully factorized prior and proposed a generalized divisive normalization (GDN) to reduce the statistical dependencies between filters. Despite GDN having managed to decrease some dependencies, there are still highly structured dependencies in the latent representation that could not be captured by the fully factorized prior. Therefore, Ballé et al. [18] improved their model with a scale hyperprior; since then, the development of deep learning-based image compression methods has become increasingly influenced by these compression frameworks [19], [20], [21], [22].

Although most existing deep learning-based compression methods are associated with RGB images, there are also many deep learning-based methods designed for HSI compression [23], [24], [25], [26], [27], [28]. La Grassa et al. [25] proposed a spectral signal compressor based on the distribution of the spectral signal source via the binary cross-entropy (BCE) loss to merge the spectral properties into the latent representation. Guo et al. [26] replaced the commonly used Gaussian distribution in RGB entropy modeling with a student's T distribution to allow the learned entropy model of the latent representation to be much closer to the actual distribution, thereby making more efficient use of the bit rates. Deng et al. [27] combined generative adversarial theory into compression to enrich the latent representation to generate more details in hyperspectral reconstruction. In reality, all these methods seem to attempt to learn a reasonably sized representation of images by incorporating a powerful predictive ability when recovering compressed images. The question then becomes what makes one latent representation better than another at a close bit rate?

In general, a good latent representation is one that can separate the distinct but informative factors of the input [29]. From the perspective of the rate–distortion trade-off, the learned features in the latent representation should tend to be well separated instead of overlapping to ensure a more compact latent representation in the condition of minimal entropy. However, this trade-off may come at the expense of collapsing or losing some informative attributes at high compression factors, which significantly degrades the performance of downstream hyperspectral tasks (e.g., a loss of category or texture). Two plausible factors include the following. First, the spectral curves of HSIs have inhomogeneity in the spatial distribution due to the influence of the imaging environment, light, or weather. That is, the gap between two different objects may decrease, resulting in a distortion of the spectral information that determines the nature of the species. Take category attribute as an example, which is shown in Fig. 1(a). The gap between similar objects, for instance, the second category and the sixth category on the Pavia University dataset (described by corresponding spectral curves, where different colors and shapes denote different categories), is small, which may make these two categories indistinguishable. Second, those attributes with indistinct differences will become constant (e.g., the red dot produced by the second category and the sixth category) after encoding and quantitation in the latent space, as shown in Fig. 1(b). As a result, these issues can cause attribute collapses and losses, thereby leading to a low-quality reconstruction due to inadequate informative factors, since we have little control over what attributes would be discarded after quantitation. In conclusion, a good representation is expressive, indicating that a learned representation should capture as many complementary explanatory factors of the input as possible. Consequently, we need to enlarge the distances across attributes to generate a discriminative latent representation and retrieve lost information as much as possible.

In this work, we propose a novel hyperspectral compression network via contrastive learning (HCCNet) to boost the latent representation and recover the lost informative attributes as much as possible. More precisely, in terms of the attribute collapses, we design a contrastive informative feature encoding (CIFE) that keeps the irrelevant features invariant and separates the relevant features by enlarging the differences over latents located in different channels of the latent representation from the positive and negative sample pair perspective. In terms of informative attribute loss, we define a contrastive-invariant feature recovery (CIFR) that seeks to maximize the mutual information between the original HSI and the reconstructed one from the feature level. Our contributions to this work can be summarized as follows.

1) We propose HCCNet, a self-supervision compression architecture via contrastive learning. This method provides an effective way of dealing with informative attribute collapses and losses that the latent representation suffered after quantification.
2) The designed CIFE enables the latent representation to extract and organize discriminative attributes from the original HSIs by enlarging the differences over latents located in different channels from the positive and negative sample pair perspective.
3) The designed CIFR enables the decoder to discover the lost attributes by maximizing the mutual information between the original HSI and the reconstructed one via contrastive feature refinement.
4) The proposed HCCNet is extensively validated on five public HSI datasets. Experimental results reveal that the HCCNet can effectively lessen the attribute collapse and loss issues in the latent representation, and outperform other state-of-the-art HSI compression methods.

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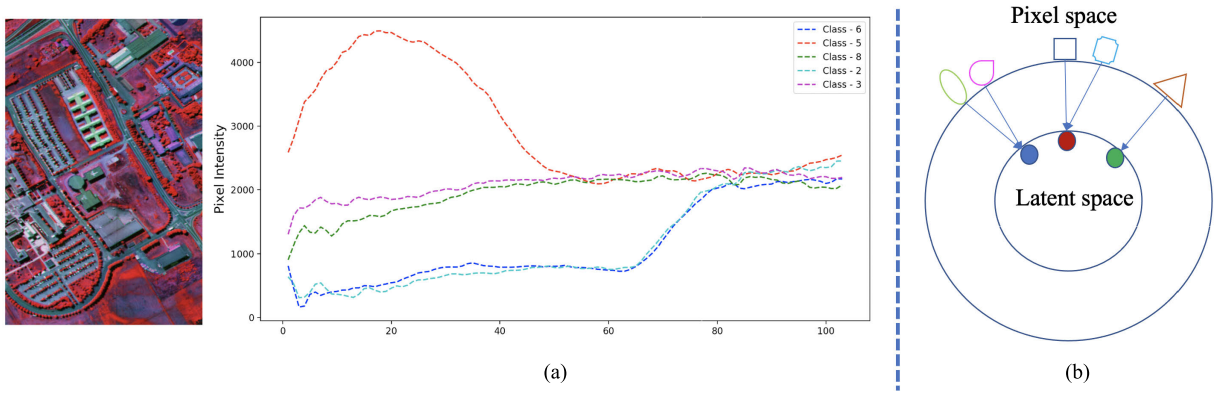


Fig. 1. Illustration of attribute collapse and loss problems. (a) Spectral curves of Pavia University. (b) Attribute collapse/loss.

The remainder of this article is organized as follows. The related work is surveyed in Section II. The proposed HCCNet is introduced in Section III. Section IV analyzes the rate–distortion performance of HCCNet; and finally, the conclusions are stated in Section V.

II. Related Work

In this section, we review the major developments in deep learning-based methods for HSI compression and works concerning contrastive representation learning.

A. Deep Learning-Based Methods for HSI Compression

Recent deep learning-based methods for HSI compression have attracted extensive attention [23], [24], [25], [26], [27], [28]. Some work focuses on designing an effective spatial–spectral representation framework to address highly correlated dependencies [24], [25]. Some efforts emphasize narrowing the difference between the learned entropy model and the actual distribution of the latent representation [26]. Some have been devoted to improve the perceptual quality of the reconstruction [27]. However, it seems unlikely to achieve a satisfactory rate–distortion trade-off when solely considering learning a representation using the minimal number of bit rates and incurring minimal pixel distortion, regardless of the limitations of HSI itself. This is because the information gap between two different objects can decrease due to different imaging conditions. Quantitation in lossy compression will accelerate this issue, which may result in missing critical characteristics during reconstruction.

B. Contrastive Learning

Contrastive learning is a self-supervision method and has been extensively discussed in unsupervised representation learning [30], [31], [32], [33], [34], [35]. The central idea is to pull the similar samples closer and push away the unrelated ones in order to keep invariants between the positive counterparts and separate the underlying variations of negatives in the representation. The definition of positive and negative counterparts is flexible in terms of the specific tasks. Recently, contrastive learning has been applied to many low-level tasks, such as image dehazing [36], image enhancement [35], and image super-resolution [37], and has achieved remarkable results. From the perspective of representation learning, the question of HSI compression can also be explained as unsupervised representation learning, which has not been explored by contrastive learning. In this article, we use contrastive learning to enlarge the discrimination over the learned latents in different channel indexes and enrich the reconstruction details by combining feature refinement with rate–distortion optimization.

III. Hyperspectral CompNet With Cross-Channel Contrastive Learning

In this section, the proposed HCCNet and its two associated components, CIFE and CIFR, are presented in detail. First, we start with an overview of HCCNet; then, descriptions of CIFE and CIFR follow.

A. Overview

As shown in Fig. 2, there are two networks in the proposed HCCNet, which will be presented in detail in Section III-B. The first one on the top is a standard compression network (CompNet), and the second one on the bottom is a pretext network (PretextNet) used to extract useful information from HSIs. Indeed, the proposed HCCNet handles HSI compression from the contrastive learning perspective via two modules: CIFE and CIFR. CIFE is used to keep the irrelevant features invariant and disentangle the relevant components of the latent representation by constructing the contrastive learning from the positive and negative sample pair perspective between latent representation Y and the features of the original HSI Y_X. More details of CIFE are presented in Section III-B. In general, we can obtain a discriminative latent representation Y after CIFE. However, the latent representation Y at this time may still suffer attribute loss if we only employ rate–distortion optimization during model learning. This is because attributes with numerous training shots may dominate the rate–distortion optimization in the pixel domain. Therefore, we design the CIFR module to restrict the optimization from the feature level to mitigate this issue. Specifically, we construct another two contrastive learning pairs to achieve feature alignment, which takes the latent representation Y, the features Y_X and

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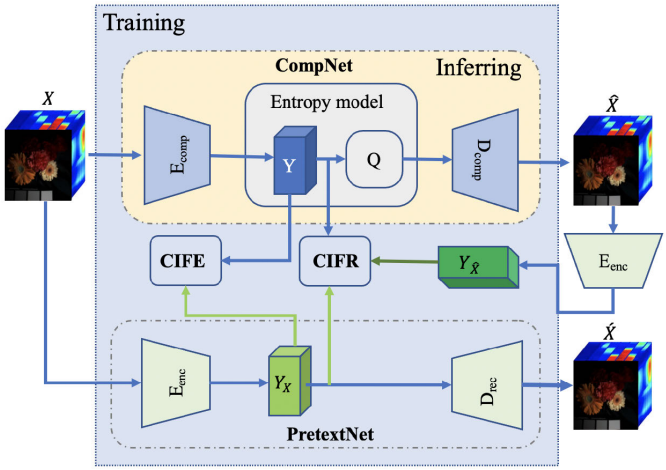


Fig. 2. Framework of the proposed HCCNet.

$Y_{\hat{X}}$ of the original, and the reconstructed HSIs as inputs. More details of CIFR are displayed in Section III-C. Note that the PretextNet, CIFE, and CIFR modules only participate in training (highlighted in light blue via a square in Fig. 2). Once inferring (highlighted in light yellow via a rounded rectangle in Fig. 2), only the CompNet is used for HSI compression.

B. Framework of HCCNet

1) CompNet: Deep learning-based compression methods provide insight into how to think about transform coding from the nonlinear perspective. Among numerous deep learning-based image compression approaches, variational image compression combined with a mean-scale hyperprior is the most widely used one [18]. To make it adaptable to HSIs and improve transform capacity, we revise the input and related convolutions. An overview of the CompNet is shown in Fig. 3. There are two autoencoders in CompNet. The encoder (labeled E_{comp} or g_a in Fig. 3) and decoder (labeled D_{comp} or g_s in Fig. 3) modules of the first autoencoder are distributed separately on the left-hand and right-hand sides, between which lie the quantization (simplified to Q in Fig. 3), arithmetic encoder and decoder (simplified to AE and AD in Fig. 3), and the second autoencoder (named hyperprior in Fig. 3), where the corresponding hyperencoder and hyperdecoder are named HE (or h_a) and HD (or h_s) in Fig. 3, respectively. Both autoencoders are implemented by cascades of convolutions and activation functions where GDN and inverse GDN (IGDN) [17] are employed in the first autoencoder and leaky rectified linear unit (LeakyReLU) in the second autoencoder. Moreover, we introduce residual blocks in the main autoencoder to enlarge the receptive field of CompNet. For hyperprior, we adopt the architecture proposed in [18].

In general, an HSI is typically modeled as $X \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the height, the width, and the number of spectral bands, respectively. After encoding, the HSI is transformed into an appropriately sized latent representation Y. The discrete-valued $\hat{Y}$ is produced by quantization, which will then be encoded into a bit stream via entropy coding (such as arithmetic coding [38]) for storage and transmission. In this case, entropy coding is always associated with a posterior distribution of the latents. In this article, we also use a mean-scale hyperprior followed [19] to approximate this probability. Certainly, we can also recover the HSI $\hat{X}$ via the decoder D_{comp} if needed. During training, the CompNet optimization can be viewed as a rate–distortion problem

$$\begin{aligned}\mathcal{L}_{\text{CompNet}} &= R + \lambda \cdot D \\ &= \underbrace{\mathbb{E}_{X\sim p_X}\left[-\log_2 p_{\hat{Y}}(Q(Y))\right]}_{\text{rate (latents)}} \\ &\quad + \underbrace{\mathbb{E}_{X\sim p_X}\left[-\log_2 p_{\hat{z}}(\hat{z})\right]}_{\text{rate (hyperlatents)}} + \lambda \cdot \underbrace{\mathbb{E}_{X\sim p_X}[d(X,\hat{X})]}_{\text{distortion}}\end{aligned} \tag{1}$$

where R and D denote the rate and distortion, respectively, and λ is a Lagrange multiplier used as a trade-off between these two terms. Specifically, the latent representation $Y = g_a(X)$ is generated by the encoder, $\hat{Y} = Q(Y)$, and the quantization Q is employed by an additive uniform noise as [18] during training. This makes the whole model be optimized with the stochastic gradient descent, since rounding to the nearest integer (used only in inference) yields zero gradients almost everywhere. $Z = h_a(Y)$ is the hyperlatent representation obtained by the hyperencoder HE, p_X represents the unknown distribution of the input HSI, $p_{\hat{Y}}$ is the learned entropy model, which is often modeled as a conditional Gaussian [the mean and scale are produced by the hyperdecoder HD, i.e., $(\mu, \sigma^2) = h_s(\hat{Z})$] convolved with a standard uniform followed [18], and $p(\hat{Z})$ is the prior of hyperlatents, which is generally modeled as a nonparametric, fully factorized distribution. The first two terms after the second equal sign are equivalent to the total encoding bit rates. Ideally, if p_X and $p_{\hat{Y}}$ are identical, the total bit rates can reach the minimum. The term after λ is distortion, where $d(\cdot)$ represents the distortion metric [e.g., mean-squared error (MSE)].

2) PretextNet: Pretext tasks, also called auxiliary tasks, are typically associated with self-supervised learning (SSL), which is a way to learn representations with automatically created pseudolabels instead of by manual annotation. In this method, we use a PretextNet to help the compression task obtain discriminative latent representation. Different from traditional pretext tasks, the designed PretextNet in this article is used to distinguish different types of attributes in the latent representation via contrastive learning. Specifically, PretextNet is used to generate features of the original and reconstructed HSIs, which are used to construct the contrastive learning condition.

The overall architecture of PretextNet is based on an autoencoder. The details of each layer are shown in Table I, where E_{enc} denotes the encoder module, which consists of two convolution layers, placed in the first and last layers and two residual blocks (named ResBS) with stride 2, D_{rec} is the decoder module that functions as an inverse process of the encoder, Conv represents convolution, the numbers followed are the kernel size, the number of filter channels, and the stride step (e.g., the convolution in the E_{enc} adopts

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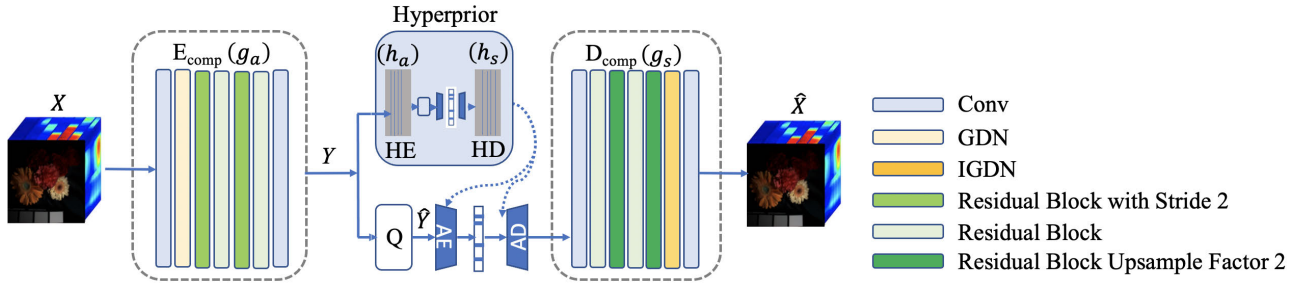


Fig. 3. Architecture of the proposed CompNet.

TABLE I
DETAILS OF THE PRETEXTNET

	Input HSI		
E_{enc}	Conv: 3×3 −192−2	BN	ReLU
	ResBS: 192−192−2	BN	ReLU
	ResBS: 192−192−2	BN	ReLU
	Conv: 3×3−320−2		
D_{rec}	Conv: 3×3 −192−2	BN	ReLU
	ResBS: 192−192−2	BN	ReLU
	ResBS: 192−192−2	BN	ReLU
	Conv: 3×3−Spectral band number−2		

192 filters with 3 × 3 kernels and stride 2), BN denotes the batch normalization, ReLU represents the activation function, and the spectral channel is the number of spectral bands of input HSIs. During producing features of the original and reconstructed HSIs, the HSI X can be transformed into a latent representation via E_{enc}. Then, with the help of D_{rec}, the HSI $\acute{X}$ can be recovered.

C. CIFE

It is well understood that a discriminative and informative latent representation helps HSI compression. The proposed CIFE module is designed specifically to explore such latent representation. First, CompNet and PretextNet create the required operating condition of CIFE, where CompNet is used to produce the latent representation Y and PretextNet is used to yield features Y_X of the original HSI without quantization. Note that both E_{comp} and E_{enc} downsample the HSI by the same factors, so that their latent representations are of identical size. Then, a cross-channel contrastive loss $\mathcal{L}_{\text{ECL}}$ during encoding is designed to improve the discrimination of latent representation Y by comparing it with Y_X from the positive and negative sample pair perspectives.

Different from traditional contrastive learning methods that employ augmented views of input as positive and negative sample pairs, we take all of the components within the same channel index as positives or negatives. Specifically, we choose each channel of the latent representation Y as the anchor, and the channel in Y_X that has the same index as the anchor is positive. Once the positive channel is defined, the remaining channels of Y_X are regarded as negative samples due to the internal image statistics [37], [39]. The schematic of this process is shown in Fig. 4. More simply, we assume that there

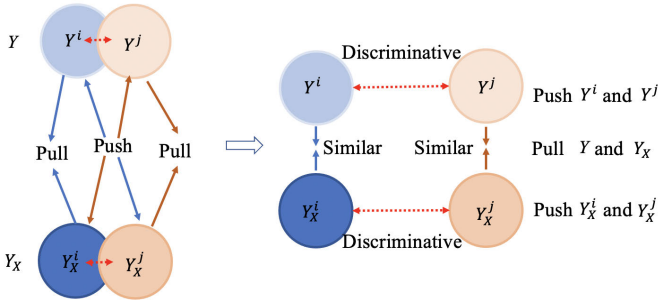


Fig. 4. Schematic when performing the encoding with cross-channel contrastive loss. Suppose that there are two features located in different channels (i and j) of the latent representation Y and Y_X, respectively. These features are colored in different shades of blue and orange to distinguish Y and Y_X, since they are similar but not identical.

are two features (shown as two circles in Fig. 4) in the latent representations Y and original features Y_X, respectively. These features are colored in different shades of blue and orange to distinguish Y and Y_X, since they are similar but not identical. Intuitively, it is clear that we can disentangle the overlapped features by enlarging (pushing away) the difference among features within the latent representation Y (also the same for Y_X), thereby obtaining a discriminative and informative latent representation. In this case, we must reduce (pull in) the difference between the latent representation Y and its corresponding channel features in Y_X simultaneously so as to keep their similarities, since Y and Y_X are analogous yet extracted by different encoders.

Based on the previous definition of positive and negative counterparts, without loss of generality, we assume that there are C channels in the latent representation Y (also the same as Y_X). Therefore, within the noise contrastive estimation framework [32], $\mathcal{L}_{ECL}(Y, Y_X)$ can be formulated as follows:

$$\mathcal{L}_{ECL}(Y, Y_X) = -\frac{1}{C}\sum_{i=1}^{C}\log\left[\frac{e^{\frac{Y^i \cdot Y_X^i}{\tau}}}{e^{\frac{Y^i \cdot Y_X^i}{\tau}} + \sum_{j\neq i}^{C} e^{\frac{Y^i \cdot Y_X^j}{\tau}}}\right] \tag{2}$$

where i and j denote the ith and jth channels, respectively, the point "·" is used to compute the dot product of these two latent vectors, and τ represents a hyperparameter (named temperature) that is used to control the shape of this dot product. Note that all latents are ℓ_2-normalized for stable training.

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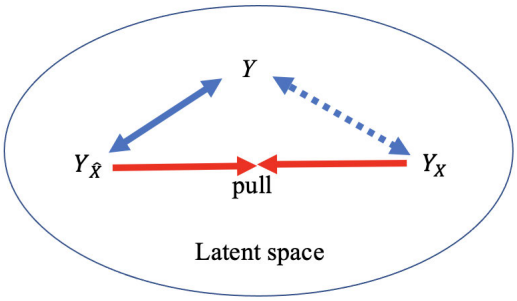


Fig. 5. Relationship of the latent representation Y, the original HSI feature Y_X, and the reconstructed HSI feature $Y_{\hat{X}}$ via contrastive learning.

D. CIFR

Directly relying on rate–distortion (e.g., MSE from pixel level) optimization to recover the lost attributes is challenging, as attributes with numerous training shots may dominate parameter learning. Therefore, we attempt to introduce comparisons between attributes in the latent space. CIFR is one such module that helps moderate this issue. Within the contrastive compression framework, we aim to improve the feature refinement ability of decoder D_{comp}. There are three components in CIFR, as shown in Fig. 2, the latent representation Y yielded by encoder E_{comp}, the original HSI feature Y_X generated by encoder E_{enc}, and the reconstructed HSI feature $Y_{\hat{X}}$ produced by the same encoder E_{enc}. These three latent representations construct a relationship, as illustrated in Fig. 5, where the blue dotted line denotes that we attempt to generate a discriminative latent representation Y via the CIFE module introduced in Section III-C, the blue line shows that we should first disentangle the relevant features of the reconstructed HSI before attribute aligning, and the red lines represent that the attributes can be refined by pulling the features of the original HSI and the reconstructed one close in the latent space. In this case, these three latent representations form a chain $Y_X \leftrightarrow X \leftrightarrow Y \leftrightarrow \widehat{X} \leftrightarrow Y_{\hat{X}}$, leading to $p(Y_X \mid X) = p(Y \mid X) = p(Y_{\hat{X}} \mid \hat{X})$. That is, these three latent representations will become closer to one another after optimization, ultimately. In these processes, latent representation Y is used as an agent to deliver discriminative features between the original HSI and the reconstructed one, which is helpful when recovering lost attributes during aligning. The alignment operation between features Y_X and $Y_{\hat{X}}$ is used to correct errors caused by the decoding process.

In the condition of the cross-channel contrastive loss $\mathcal{L}_{ECL}(Y, Y_X)$ designed in CIFE, we define another two during decoding. Take $\mathcal{L}_{RCL}(Y, Y_{\hat{X}})$ as an example, where each of the C channels of the latent representation Y serves as the positive, the corresponding channel in $Y_{\hat{X}}$ serves as the anchor, and the remaining $(C-1)$ channels of Y serve as negatives. Consequently, $\mathcal{L}_{RCL}(Y, Y_{\hat{X}})$ can be formulated as follows:

$$\mathcal{L}_{RCL}(Y, Y_{\hat{X}}) = -\frac{1}{C}\sum_{i=1}^{C} \log\left[\frac{e^{\frac{Y^i \cdot Y_{\hat{X}}^i}{\tau}}}{e^{\frac{Y^i \cdot Y_{\hat{X}}^i}{\tau}} + \sum_{j\neq i}^{C} e^{\frac{Y^i \cdot Y_{\hat{X}}^j}{\tau}}}\right]. \quad (3)$$

The form of (3) is similar to (2) but with a different intended purpose in that we aim to generate a discriminative representation $Y_{\hat{X}}$ before the attribute aligns with assistance from latent representation Y, which is shown in Fig. 6. This helps to maximize the amount of information, since the underlying idea is the maximum entropy principle. In addition, $\mathcal{L}_{RCL}(Y_X, Y_{\hat{X}})$ is used to align attributes from the feature level in the latent space to compensate for the weakness that attributes with numerous training shots may come to dominate the rate–distortion optimization in the pixel domain. This also provides insight into how to avoid information distortion from a contrastive learning perspective by maximizing mutual information between the original HSI and the reconstructed one, which is shown in Fig. 6 (right).

Therefore, the loss function of the HCCNet via cross-channel contrastive learning can be formulated by

$$\mathcal{L}_{\text{HCCNet}} = \mathcal{L}_{ECL}(Y, Y_X) + \mathcal{L}_{\text{RCL}}(Y, Y_{\hat{X}}) + \mathcal{L}_{\text{RCL}}(Y_X, Y_{\hat{X}}) + \mathcal{L}_{\text{CompNet}}. \quad (4)$$

Note that there are two phases of the proposed HCCNet during training. The first one is a pretraining process, which is used to obtain a stable PretextNet via a ℓ_1 loss regardless of CompNet. In this phase, CompNet also participates in training, which can provide a good initial value. In this case, each bit rate corresponds to a PretextNet parameter (i.e., the PretextNet needs to be retrained for different bit rates). Then, we freeze the PretextNet (i.e., the PretextNet has no gradient propagation in the following phase) and continue the second phase with (4).

Consequently, the cross-channel contrastive losses enable the encoder E_{comp} to obtain a discriminative latent representation Y, regardless of the influence of the small gap between objects caused by the imaging conditions, and help the decoder D_{comp} capture most of the lost attributes at a fixed bit rate.

IV. Experimental Results and Analysis

The proposed HCCNet is verified on five datasets involving four public remote sensing HSI datasets (i.e., Chikusei [40], Washington DC, Houston (DFC2018), and Pavia Centre) and a natural HSI (i.e., CAVE [41]) to demonstrate its superior compression potential. Furthermore, we also compare it with the existing commonly used HSI compression methods involving three handcrafted approaches (i.e., KLT + JPEG2000 [12], 3-D-Speck [42], and enhanced multivariance products representation (EMPR) [43]) and five state-of-the-art deep-learning-based approaches (i.e., mean scale [18], context [19], Cheng2020att [21], high-order MRF-oriented attention (HMA) [44], and checkerboard context model (CCM) [45]).

A. Dataset

1) Chikusei: The Chikusei dataset captured by the Headwall Hyperspec-VNIR-C imaging sensor is about the agricultural and urban areas of Chikusei. It has a spatial size of 2517 × 2335 with 128 spectral bands involving a wavelength of 363–1018 nm. To intuitively understand this dataset, we display its false color in Fig. 7(a).

2) Washington DC: The Washington DC is airborne hyperspectral data captured by the Spectral Information Technology Application Center, VA. It has a spatial size of

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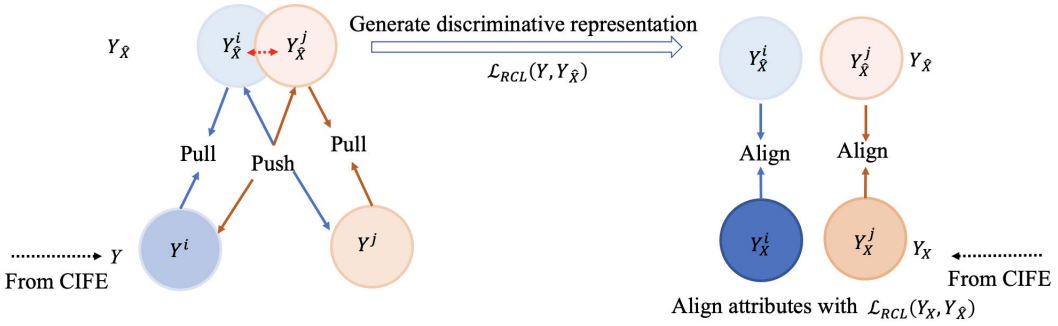


Fig. 6. Schematic of the overall CIFR process. Suppose that there are two features located in different channels (i and j) of the representation $Y_{\hat{X}}$, colored with lighter blue and orange than the representation Y, respectively.

1280 × 307 with 191 spectral bands, covering a wavelength of 400–2400 nm. In addition, we display its false color in Fig. 7(b).

3) Houston: There are two datasets of Houston scenes, and we utilize the DFC2018 dataset in our experiments. This dataset was captured in 2018 by the Compact Airborne Spectrographic Imager. It has a spatial size of 1202 × 4172 with 50 spectral bands that cover a wavelength range of 380–1050 nm. In addition, we display its false color in Fig. 7(c).

4) Pavia Centre: The Pavia Centre dataset is captured by the Reflective Optics System Imaging Spectrometer sensor, which has a spatial size of 1096 × 715 with 102 spectral bands. There are nine classes in this dataset. In addition, we display its false color in Fig. 7(d).

5) CAVE: CAVE is a natural HSI dataset captured by the generalized assorted pixel (GAP) camera, covering a wavelength of 400–700 nm. There are 32 scenes in this dataset involving a large number of objects and materials, including skin and both real and fake fruits, where each individual scene has a spatial size of 512 × 512 with 31 spectral bands. A representative thumbnail is shown in Fig. 7(e) in false color.

B. Evaluation Metrics

Certainly, evaluation metrics are necessary when drawing rate–distortion curves. In this article, we use bit per pixel per band (bpppb) to determine the degree of compression for the HSIs. Meanwhile, the peak signal-to-noise ratio (PSNR), multiscale structural similarity metric (MS-SSIM) [46], and spectral angle mapping (SAM) [47] are used to quantitatively measure the distortion.

1) PSNR: PSNR is a widely used metric for the evaluation of image quality, and the larger its value, the better the image quality becomes. Unlike the traditional PSNR calculation method, we take the maximum pixel in the ith band as the peak signal, following [18], rather than a fixed value, which is more appropriate for HSIs. The average PSNR for a B-spectral band HSI can be formulated by

$$\mathrm{PSNR}(X, \hat{X}) = \frac{1}{B}\sum_{i=1}^{B} 10\log_{10}\left(\frac{\max^2(\hat{X}^i)}{\mathrm{MSE}(X, \hat{X})}\right) \tag{5}$$

where $\mathrm{MSE}(X, \widehat{X}) = (1/(H \times W \times B))\|X - \widehat{X}\|_2^2$ is the MSE between the original X and the reconstructed $\widehat{X}$, $\max^2(\cdot)$ is the square of the maximum pixel in the ith band, and B denotes the spectral band number of the HSI.

2) MS-SSIM: MS-SSIM is another commonly used metric that evaluates the quality of images at different resolutions. The values of MS-SSIM are between 0 and 1, and the larger the value is, the better the quality of the image. The details of MS-SSIM can be found in [46]. Since almost all values of MS-SSIM were concentrated at a value of approximately 0.9, we convert the value d of MS-SSIM to $-10\log_{10}(1-d)$, following [18], to amplify the difference between the original HSI and the reconstructed one.

3) SAM: In addition, we also choose a distortion metric SAM to evaluate the quality of HSI in terms of the spectral dimension. Different from the two metrics that were listed previously, a small SAM value reveals a good reconstruction performance. The mathematical description can be formulated by

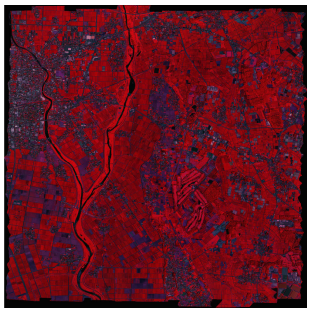
$$\mathrm{SAM}(X, \hat{X}) = \frac{1}{H \times W} \times \sum_{j=1}^{H\times W} \cos^{-1}\left(\frac{x_j \odot \hat{x}_j}{\|x_j\|\|\hat{x}_j\|}\right) \tag{6}$$

where x_j and $\hat{x}_j$ correspond to the jth pixel of the original X and the reconstructed HSI $\widehat{X}$, respectively, $\odot$ denotes inner product, and $\|\cdot\|$ is the ℓ_2 norm.

C. Setup

1) Dataset Processing: The five datasets are divided into two groups. First, the first group is a mix of HSIs (named HSImix), which is a combination of four remote sensing HSI datasets (i.e., Chikusei, Washington DC, Houston, and Pavia Centre). To simplify the training process, this group dataset, following [48], is cropped into 50 spectral bands, as this is the minimum number of spectral bands (i.e., Houston) in these four HSIs. Before training, these HSIs are then randomly cropped into nonoverlapping 256 × 256 patches, which are partitioned for training, validation, and testing, at an 8:1:1 ratio. Second, the second group is the natural HSI CAVE. There are 32 HSIs in this dataset, and we also divide these images for training, validation, and testing at an 8:1:1 ratio.

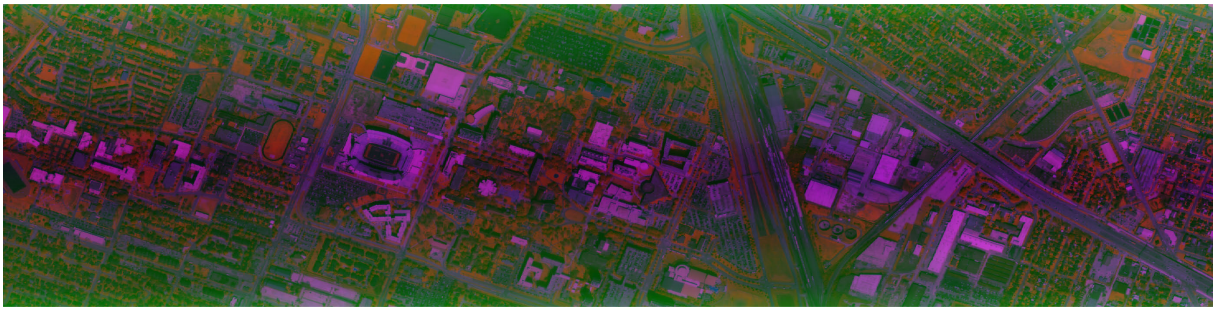
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(a) Chikusei data with bands 100, 10, and 50.



(b) Washington DC data with bands 60, 27, and 17.



(c) Houston data with bands 30, 49, and 15.



(d) Pavia Centre data with bands 100, 10, and 50.



(e) CAVE data with bands 15, 5, and 30.

Fig. 7. (a) Chikusei. (b) Washington DC. (c) Houston. (d) Pavia Centre. (e) CAVE.

2) Device and Settings: All traditional compression methods are run on the same CPU (Intel i9-10920X CPU @ 3.5 GHz) with open software or codes, where JPEG2000 is employed with Kakadu version 8.0.5, 3-D-Speck is implemented by QccPack, and EMPR is conducted via the discrete Haar transform after JPEG2000. In the case of deep learning-based compression approaches, they are all implemented with the PyTorch framework on NVIDIA GTX 3090 GPUs. In addition, these approaches are trained with the Adam optimizer, and their learning rates are all followed [18]. During training, data augmentation methods, such as random cropping, are used to enrich the training data further. All models are trained on 300 epochs, where the first 150 epochs are used to obtain a stable PretextNet. Then, we freeze the PretextNet and continue the remaining 150 epochs with rate–distortion and contrastive loss.

D. Experimental Results

Note that all deep learning-based models are trained using MSE. Except for CAVE, which has a spatial size of 512×512, the remaining visualization examples have a spatial size of 256×256.

1) Ablation Results: To evaluate the effectiveness of the proposed two modules, i.e., CIFE and CIFR, we conduct ablation experiments on two datasets, i.e., CAVE and Pavia Centre, which include two different spectral bands and different types of objects. The rate–distortion curves are shown in Figs. 8 and 9. CompNet works as the baseline,

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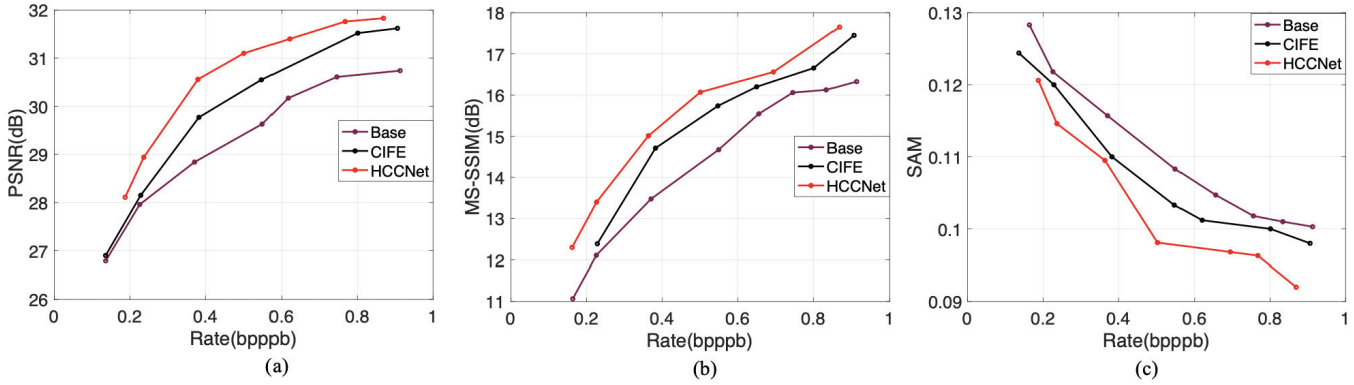


Fig. 8. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Pavia Centre dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The rate–distortion results illustrate the effectiveness of CIFE and CIFR.

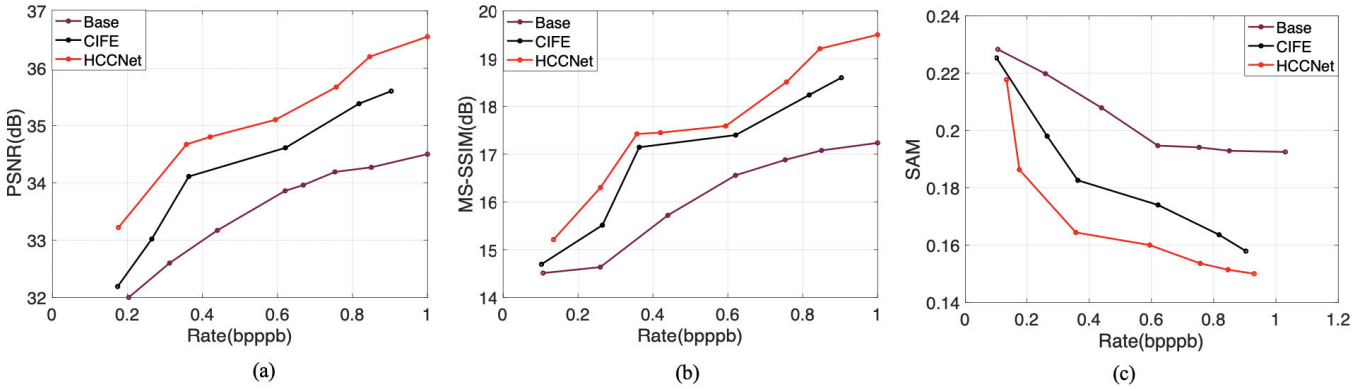


Fig. 9. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the CAVE dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The rate–distortion results illustrate the effectiveness of CIFE and CIFR.

which is named Base, as shown in the purple plots. CIFE denotes that we perform the HCCNet without CIFR, where the rate–distortion curves are plotted in black color. Finally, the rate–distortion performances of the proposed HCCNet, which involves both CIFE and CIFR modules, are plotted in red.

We find that the PSNRs are improved significantly after 0.2 bpppb on the Pavia Centre dataset when we combine CIFE with CompNet, such as from 28.84 dB (at 0.3709 bpppb) to 29.77 dB (at 0.3726 bpppb). In the case of MS-SSIMs and SAMs, the average performances increased by approximately 0.9 dB and decreased by approximately 4.2%, respectively. These results verified that CompNet can produce a good representation that benefited the compression performance via CIFE. The results on CAVE demonstrated this point again. In addition, once the CompNet included both CIFE and CIFR modules, the overall compression performances on the Pavia Centre and CAVE further improved. In conclusion, both Figs. 8 and 9 illustrate that the proposed CIFE and CIFR modules can clearly improve the performance of the compression model.

2) Compression Performance Comparisons: In this section, we compare our HCCNet with the existing commonly used HSI compression methods on five HSI datasets to illustrate its superiority. The rate–distortion performances and visual comparisons are shown from Figs. 10–19.

Fig. 10 shows the summarized RD curves on the Chikusei dataset in terms of PSNR, MS-SSIM, and SAM, which are arranged from left to right. The results on these three metrics indicate that our HCCNet outperforms the existing commonly used HSI compression methods over almost all bit rates, including either the handcrafted approaches or other state-of-the-art deep learning-based methods. Moreover, the quantitative results and corresponding visual comparisons at close bit rates in Fig. 11 illustrate this point again. Note that the quantitative results and corresponding visual comparisons are shown at a close bit rate, since it is difficult for most compression methods to obtain a precise target bit rate. From the visual images, we can find that there are many small objects and texture details in Chikusei, where some small buildings or textures can be easily collapsed or blurred during lossy compression at a low bit rate. However, our proposed HCCNet can effectively alleviate this issue; for instance, in this dataset, texture details of objects are relatively complete and clear, which are highlighted with a yellow ellipse. That is, contrastive learning is indeed advantageous for generating a discriminative latent representation and preserving details that benefit accurate reconstruction.

Apart from the Chikusei dataset, we also conduct extensive experiments on the other three publicly available HSI datasets, including various object types and scales, to verify the advantage of our HCCNet. Fig. 12 presents the RD curves of the Washington DC dataset. As expected, our HCCNet behaves competitively across almost all bit rates when evaluating distortion with PSNR and MS-SSIM and performs the best

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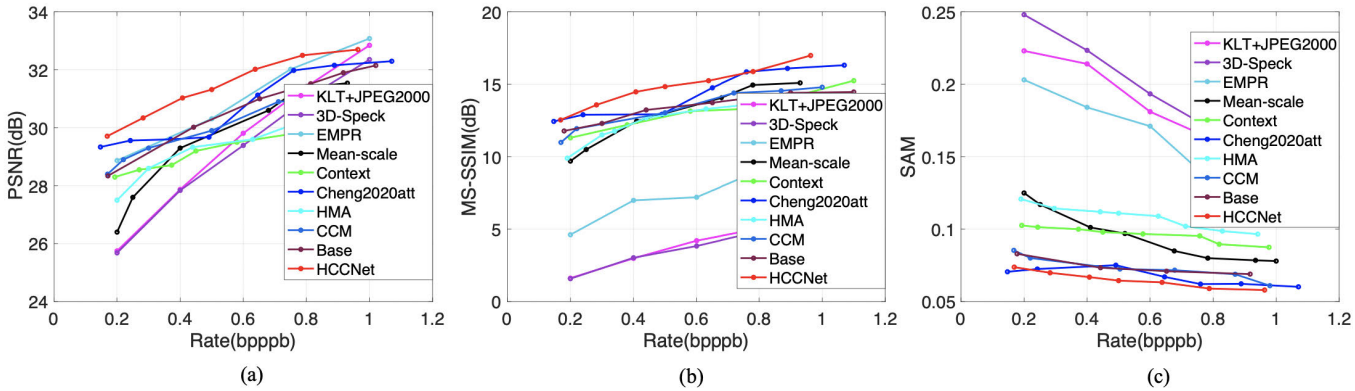


Fig. 10. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Chikusei dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The proposed HCCNet (red line) achieves the best RD performance on these three metrics compared with other methods.

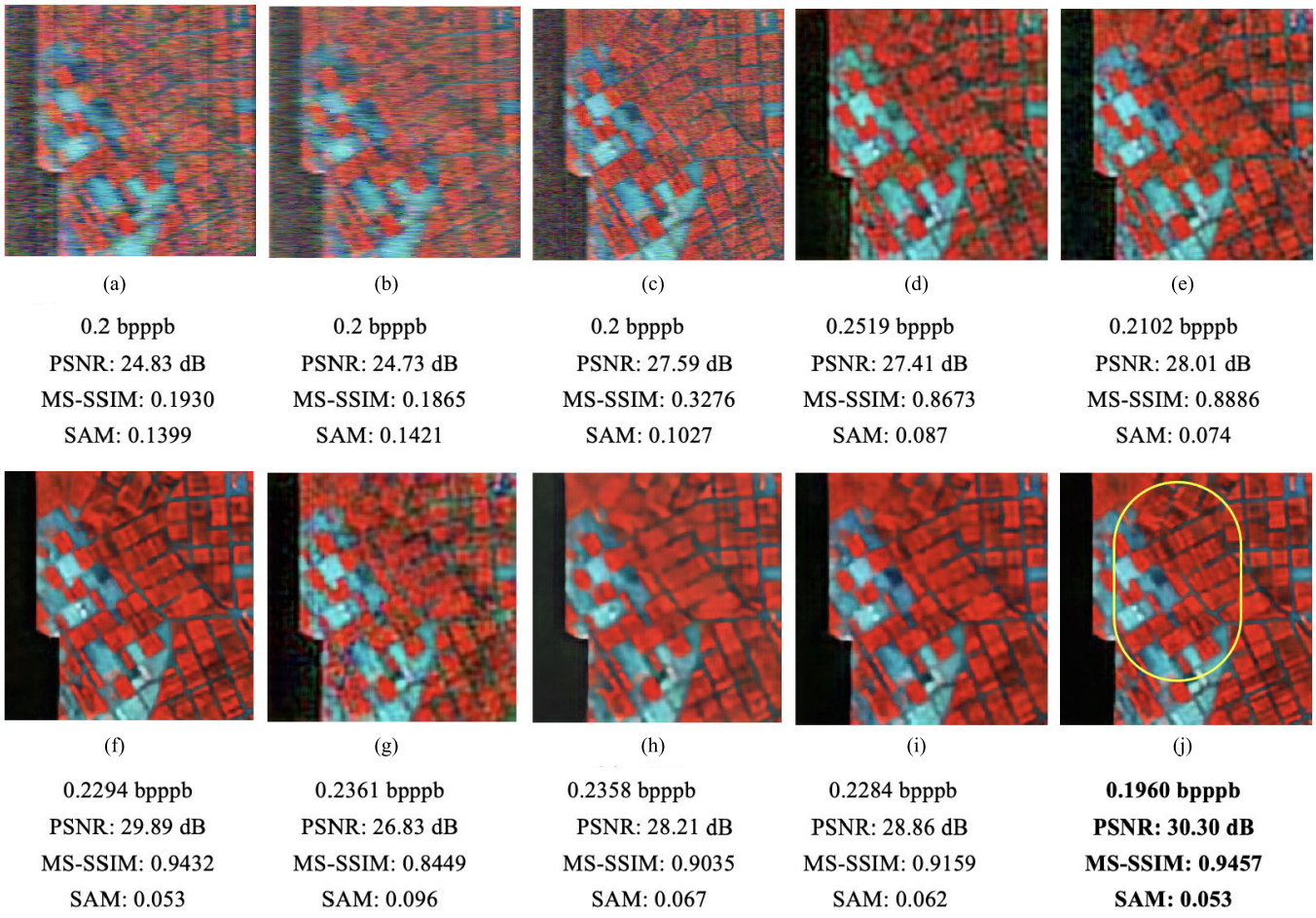


Fig. 11. Visual comparisons at close bit rates on an example patch from the Chikusei dataset. Bands 49, 5, and 20 made the false color. (a) KLT + JPEG2000. (b) 3-D-Speck. (c) EMPR. (d) Mean scale. (e) Context. (f) Cheng2020att. (g) HMA. (h) CCM. (i) Base. (j) HCCNet.

throughout all bit rates when evaluating distortion with SAM. However, the phenomenon that traditional codecs can surpass the HCCNet at high bit rates reveals one point that deep learning-based approaches can easily develop performance saturation regardless of the increasing bit rates. The reason for this issue can be the insufficient training data compared with traditional codecs, which are optimized with image-specific context during model learning. After all, there are only 1280×307 pixels in the spatial size of Washington DC, which is minimal compared with other HSI datasets. Note that CAVE is also larger than Washington DC, as there are more than one 512×512 HSI in the experiments. Fig. 13 provides a visual comparison for one of the test patches of Washington DC, where the proposed HCCNet yields clearly better visual perception than all other methods. Figs. 14 and 16 reveal the RD performances on the Houston and Pavia Centre datasets, where Figs. 15 and 17 correspond to their visual comparisons. The results also illustrate that

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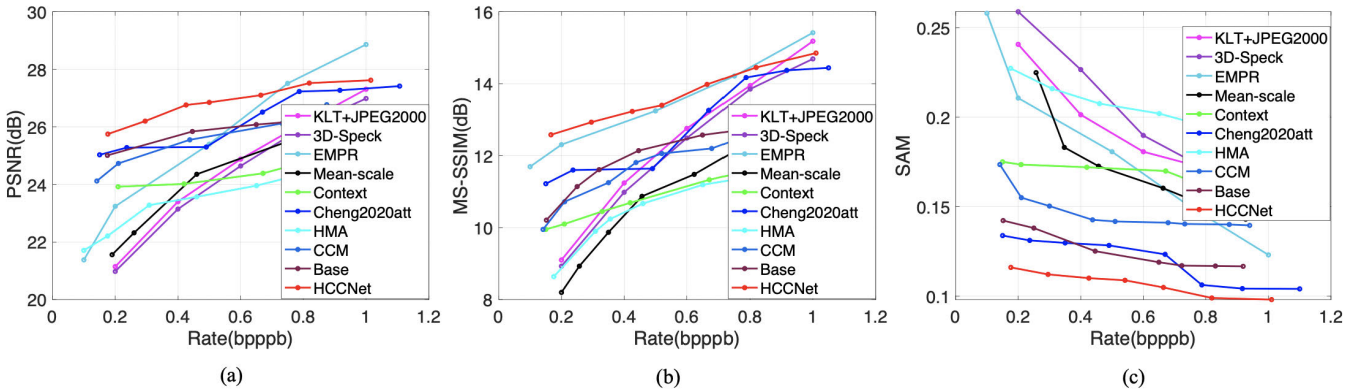


Fig. 12. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Washington DC dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The proposed HCCNet (red line) performs competitively compared with other methods.

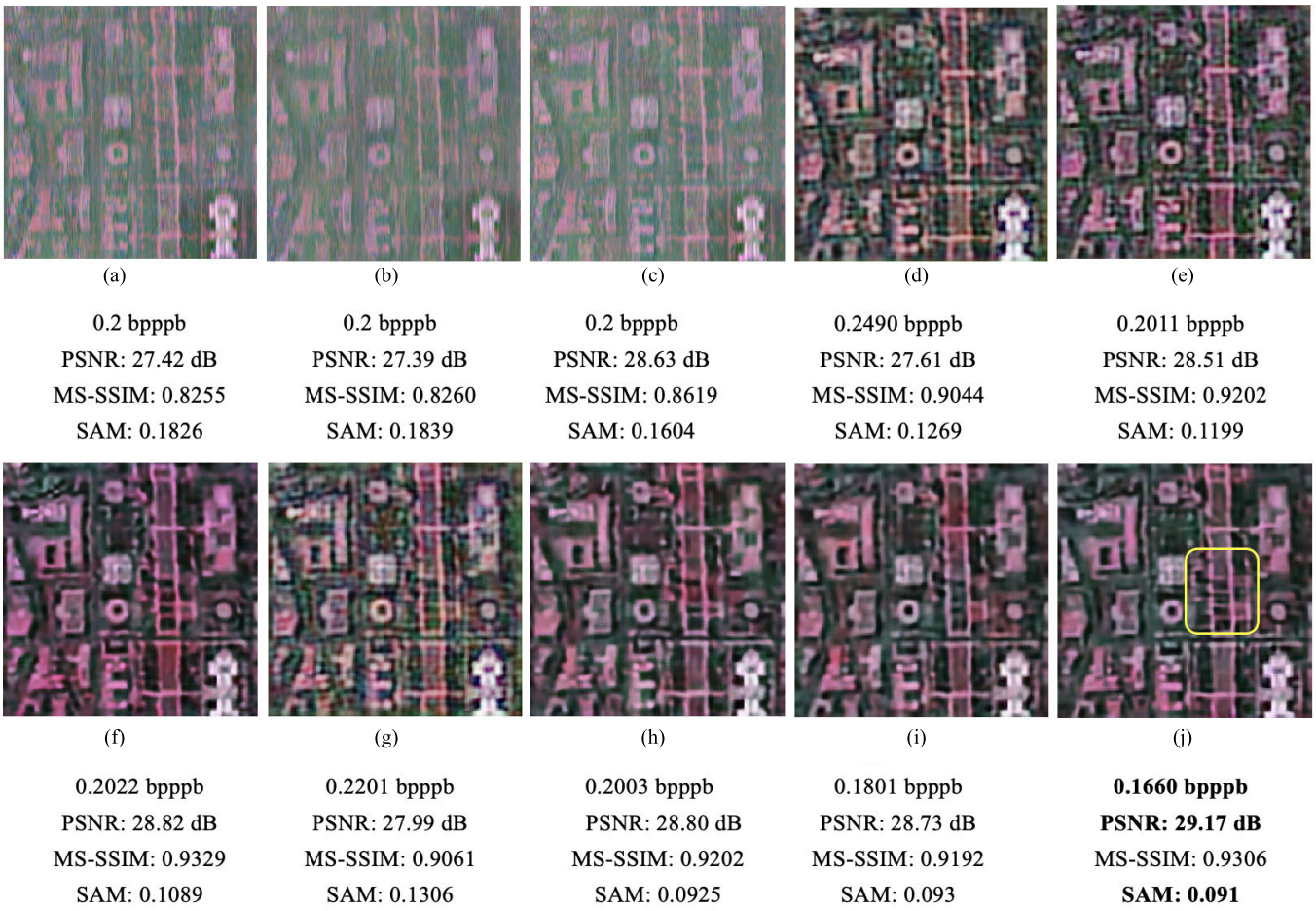


Fig. 13. Visual comparisons at close bit rates on an example patch of the Washington DC dataset. Bands 49, 5, and 20 made the false color. (a) KLT + JPEG2000. (b) 3-D-Speck. (c) EMPR. (d) Mean scale. (e) Context. (f) Cheng2020att. (g) HMA. (h) CCM. (i) Base. (j) HCCNet.

our HCCNet performs the best on these two remote sensing datasets.

In addition, we also implement our experiments on a natural HSI dataset, i.e., CAVE, which involves clearly larger objects compared with those present in the remote sensing HSIs. In this case, we can more intuitively observe the reconstructions. Fig. 18 depicts the RD performance of the CAVE dataset. From the RD curves, we can conclude that the proposed HCCNet still outperforms all the other approaches, especially in terms of SAM. For PSNR, our HCCNet is also the best, although the RD curve becomes very close to the traditional codec EMPR at high bit rates. The reason for this can also be the data-driven method when seeking the optimal model parameters. Moreover, Fig. 19 shows the visual comparisons over an individual test HSI of the CAVE dataset. From the results, we find that most codecs exhibit

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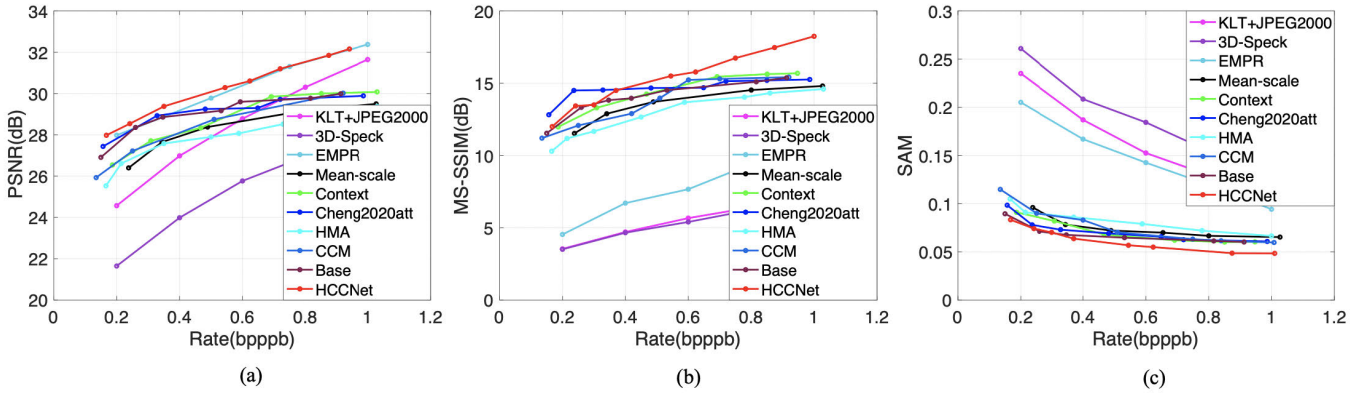


Fig. 14. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Houston dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The proposed HCCNet (red line) achieves the best RD performance on these three metrics compared with other methods.

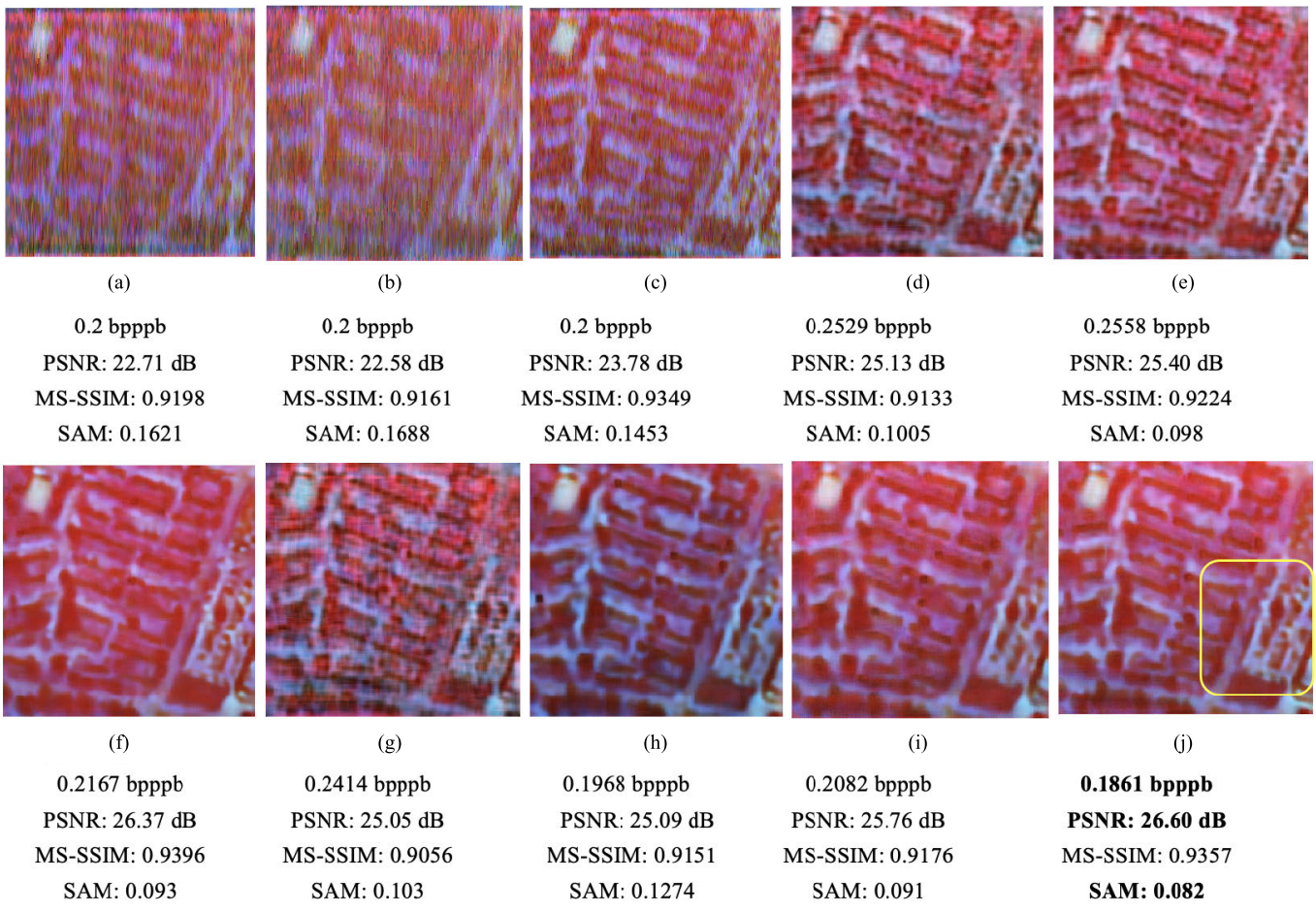


Fig. 15. Visual comparisons at close bit rates on an example patch of the Houston dataset. Bands 49, 5, and 20 made the false color. (a) KLT + JPEG2000. (b) 3-D-Speck. (c) EMPR. (d) Mean scale. (e) Context. (f) Cheng2020att. (g) HMA. (h) CCM. (i) Base. (j) HCCNet.

banding and ringing noises that surround objects even if the attributes are kept relatively complete. However, the proposed HCCNet can not only reduce the artifacts but also relieve the attribute collapse and loss problems, such as a clearly better center of the lemon slices and relatively complete strawberry leaves, which are marked by yellow squares in Fig. 19. More importantly, we have the best reconstruction quality compared with other codecs. Consequently, all compression results in this article demonstrate the performance gains of the proposed HCCNet.

3) Classification Results: As we know, there are many attributes in the latent representation Y. Apart from the compression performance, we conduct a practical application, i.e., hyperspectral classification, to demonstrate the proposed HCCNet, which is quite helpful to improve the discriminative ability of latent representation. We implement our experiments

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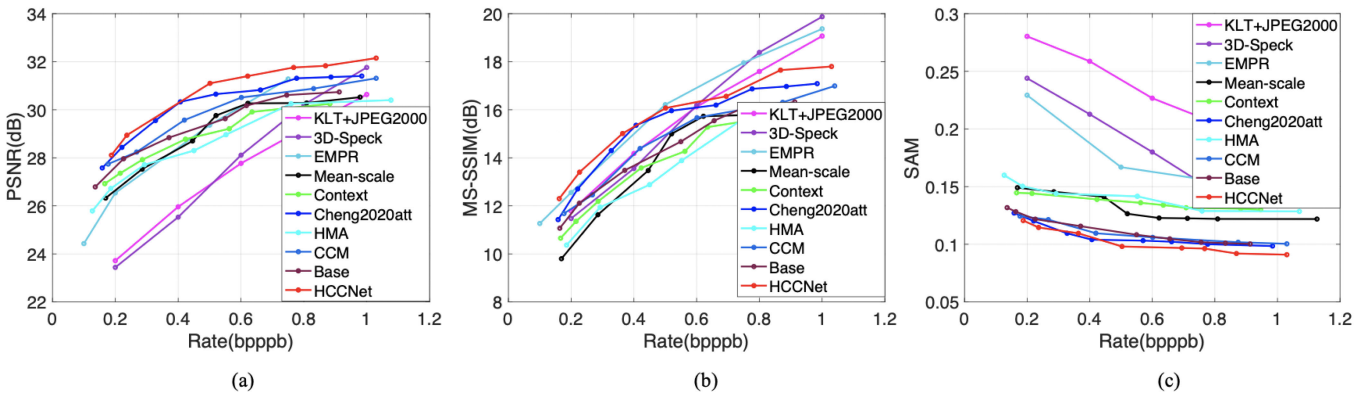


Fig. 16. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Pavia Centre dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The proposed HCCNet (red line) achieves the best RD performance on these three metrics compared with other methods.

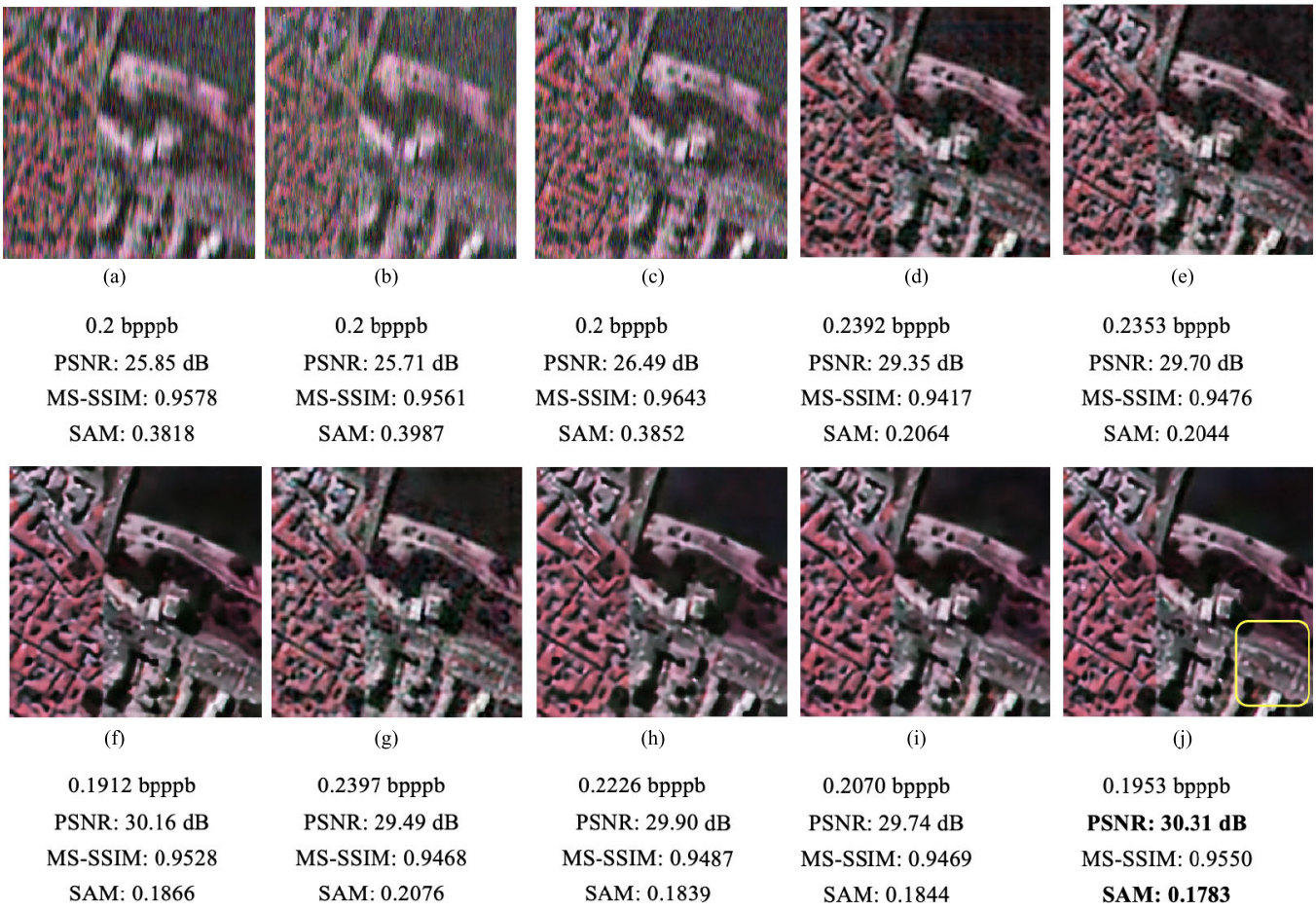


Fig. 17. Visual comparisons at close bit rates on an example patch of the Pavia Centre dataset. Bands 49, 5, and 20 made the false color. (a) KLT + JPEG2000. (b) 3-D-Speck. (c) EMPR. (d) Mean scale. (e) Context. (f) Cheng2020att. (g) HMA. (h) CCM. (i) Base. (j) HCCNet.

on the original HSIs and the reconstructed ones (on the Pavia Centre dataset across two bit rates) obtained from all compression methods via a simple but commonly used classification algorithm SVM [49]. We use principal component analysis to reduce the spectral channel into 50 when hyperspectral classification since all learned compression models are trained only for 50 spectral channels of hyperspectral remote sensing images. The overall accuracy (OA) [50], kappa coefficient (KC) [51], and class accuracy (CA) [52] are used to evaluate the performance of HSI classification, where the closer the metrics are to 1, the better the classification results.

In order to illustrate the efficacy of the proposed HCCNet intuitively, we make a visualization of the latent representation Y obtained by comparison compression methods via 2-D t-distributed stochastic neighbor embedding (t-SNE) from the category perspective. The results are shown in Fig. 20, where different colors represent different classes. From Fig. 20,

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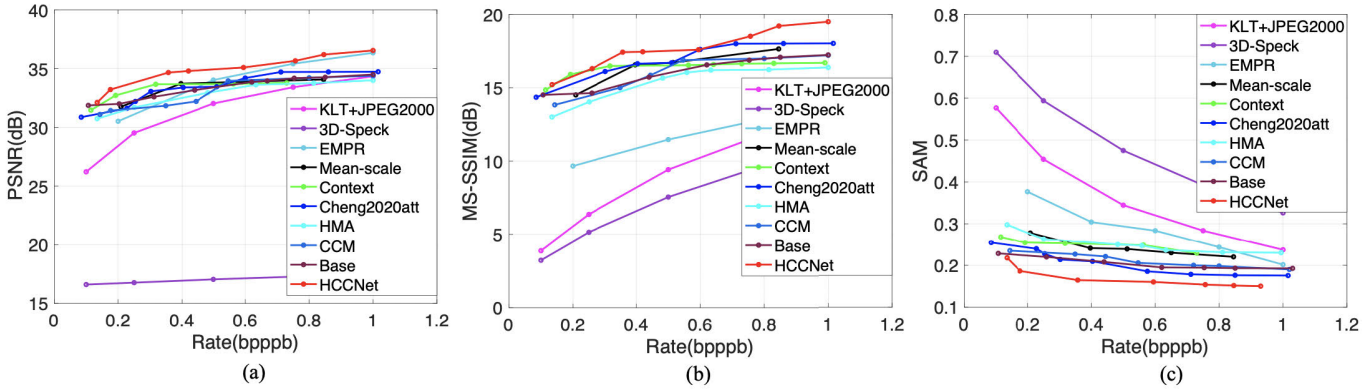


Fig. 18. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the CAVE dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The proposed HCCNet (red line) achieves the best RD performance on these three metrics compared with other methods.

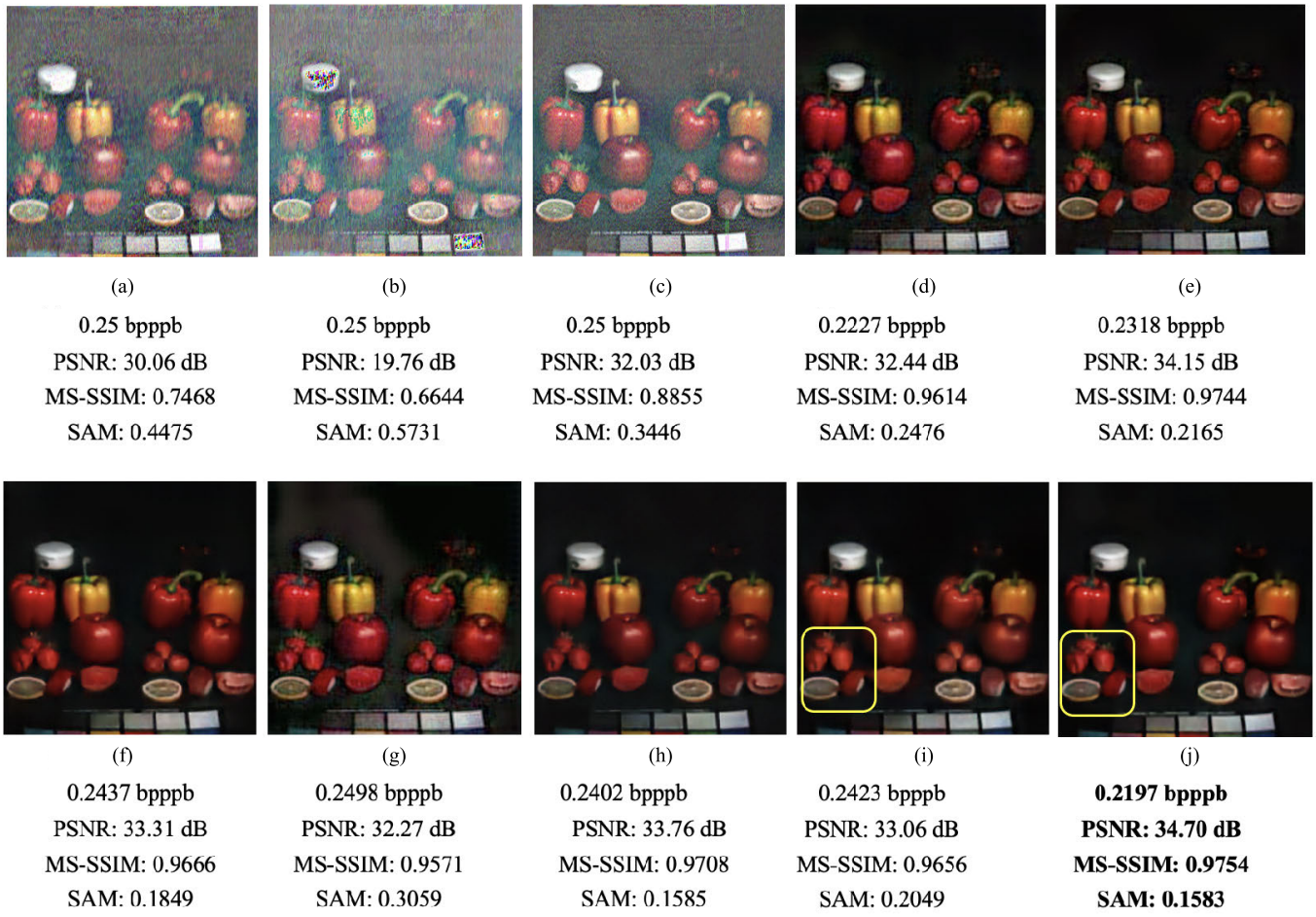


Fig. 19. Visual comparisons at close bit rates on an example image from the CAVE dataset. Bands 15, 5, and 30 made the false color. (a) KLT + JPEG2000. (b) 3-D-Speck. (c) EMPR. (d) Mean scale. (e) Context. (f) Cheng2020att. (g) HMA. (h) CCM. (i) Base. (j) HCCNet.

we can find that categories of the latent representation Y obtained by the HCCNet are distributed more separately and clearly, which illustrates that the proposed method does have the expected discriminative ability.

Moreover, the accuracy metrics of the corresponding classification results are shown in Tables II and III. From these tables, we can find that the proposed HCCNet yields state-of-the-art classification performance in terms of OA and KC, consistently across two bit rates. This demonstrates that a better reconstruction may provide a higher classification accuracy. In addition, the competitive or even superior CA illustrates that the proposed method can indeed improve the discrimination between classes.

E. Model Complexity

It is very necessary to check the model complexity of the proposed method, as it is a crucial part of the performance. We verify the complexity of the proposed HCCNet to the

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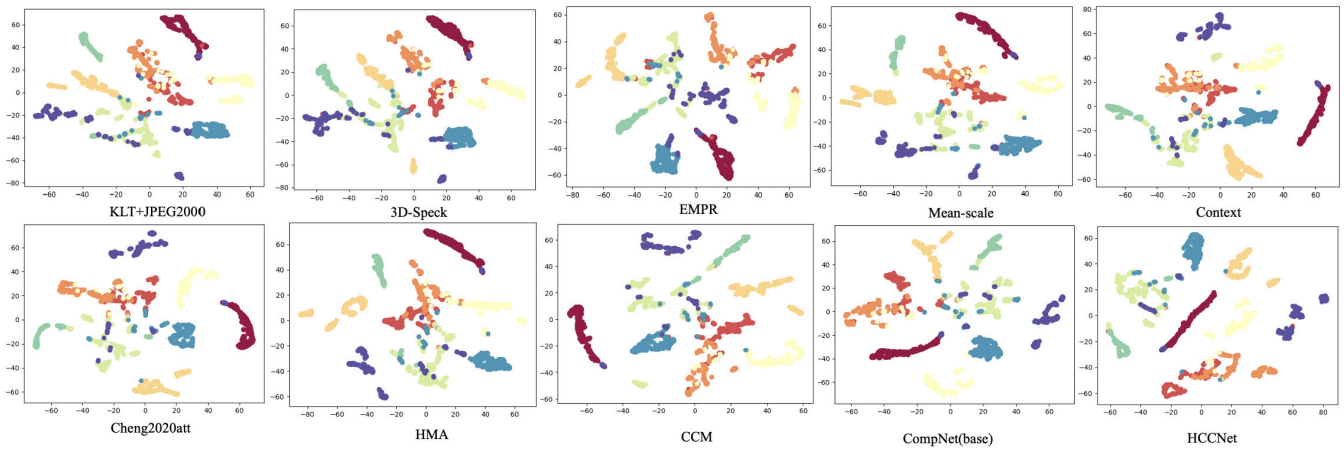


Fig. 20. 2-D t-SNE visualizations of the latent representation Y obtained by comparison compression methods on the Pavia dataset.

TABLE II

CLASSIFICATION ACCURACY OF DIFFERENT RECONSTRUCTIONS OBTAINED FROM THE COMPARED COMPRESSION METHODS AND THE PROPOSED HCCNET FOR THE PAVIA CENTRE DATASET AT AROUND 0.2 BPPPB

Class	Ori	KLT+JPEG2000	3D-Speck	EMPR	Mean-scale	Context	Cheng2020att	HMA	CCM	Base	HCCNet
1	0.999	0.983	0.986	0.987	0.986	0.991	0.999	0.987	0.997	0.997	0.998
2	0.929	0.928	0.926	0.936	0.883	0.884	0.890	0.862	0.906	0.883	0.890
3	0.816	0.798	0.807	0.841	0.640	0.644	0.662	0.577	0.731	0.661	0.666
4	0.604	0.147	0.182	0.159	0.000	0.000	0.212	0.016	0.448	0.156	**0.563**
5	0.833	0.606	0.635	0.651	0.522	0.567	0.750	0.454	0.697	0.700	0.709
6	0.895	0.719	0.719	0.795	0.741	0.758	0.820	0.772	0.798	0.779	0.815
7	0.857	0.733	0.737	0.746	0.692	0.711	0.786	0.715	0.746	0.749	**0.786**
8	0.953	0.890	0.895	0.910	0.883	0.891	0.914	0.855	0.919	0.906	**0.920**
9	0.998	0.327	0.482	0.596	0.533	0.764	0.957	0.612	0.930	0.941	**0.958**
OA	0.952	0.890	0.895	0.909	0.883	0.893	0.921	0.884	0.920	0.911	**0.924**
KC	0.932	0.841	0.849	0.868	0.830	0.845	0.887	0.832	0.885	0.872	**0.891**

1: Water 2: Trees 3: Asphalt 4: Self-Blocking Bricks 5: Bitumen 6: Tiles 7: Shadows 8: Meadows 9: Bare Soil

TABLE III

CLASSIFICATION ACCURACY OF DIFFERENT RECONSTRUCTIONS OBTAINED FROM THE COMPARED COMPRESSION METHODS AND THE PROPOSED HCCNET FOR THE PAVIA CENTRE DATASET AT AROUND 1 BPPPB

Class	Ori	KLT+JPEG2000	3D-Speck	EMPR	Mean-scale	Context	Cheng2020att	HMA	CCM	Base	HCCNet
1	0.999	0.992	0.992	0.995	0.995	0.994	0.998	0.993	0.997	0.999	**0.999**
2	0.929	0.933	0.932	0.929	0.896	0.899	0.917	0.895	0.918	0.905	0.917
3	0.816	0.833	0.832	0.818	0.746	0.750	0.794	0.737	0.789	0.744	0.777
4	0.604	0.501	0.421	0.532	0.448	0.430	0.624	0.358	0.599	0.584	**0.632**
5	0.833	0.640	0.628	0.697	0.539	0.503	0.694	0.517	0.645	0.720	**0.733**
6	0.895	0.862	0.859	0.876	0.848	0.850	0.872	0.823	0.856	0.849	0.865
7	0.857	0.817	0.817	0.831	0.807	0.812	0.818	0.773	0.808	0.793	0.805
8	0.953	0.927	0.922	0.936	0.915	0.912	0.937	0.909	0.929	0.933	**0.942**
9	0.998	0.798	0.795	0.889	0.879	0.871	0.971	0.840	0.953	0.991	**0.993**
OA	0.952	0.927	0.924	0.935	0.918	0.916	0.937	0.911	0.932	0.934	**0.940**
KC	0.932	0.896	0.891	0.907	0.883	0.880	0.911	0.871	0.902	0.905	**0.915**

1: Water 2: Trees 3: Asphalt 4: Self-Blocking Bricks 5: Bitumen 6: Tiles 7: Shadows 8: Meadows 9: Bare Soil

comparison algorithms via memory usage (measured with the number of parameters), floating point operations (FLOPs), and inferring time (including Enc-time and Dec-time). The experimental results are shown in Table II. The symbol "–" shows no results, since the parameters of traditional compression methods are manually set instead of learning. All codecs were running on the same CPU during the inference process.

Table IV shows that the proposed HCCNet is competitive in compressing various types of HSIs in terms of inference time. In the case of memory usage and FLOPs, the proposed method does expend some computer resources to assist the CompNet to capture discriminative attributes and lessen attribute loss. However, it does not matter after all we do not struggle for saving capacity of the HCCNet, as we strive to solve the attribute collapse and loss problems that occur in the latent

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TABLE IV
MODEL PARAMETERS, FLOPS, AND INFERRING TIME (INCLUDING ENC-TIME AND DEC-TIME) ON TWO GROUPS OF HSIS FOR THE PROPOSED HCCNET AND COMPARISON METHODS

Datasets	Methods	Parameters(M)	FLOPs(G)	Enc-time(s)	Dec-time(s)
CAVE	KLT+JPEG2000	–	–	2.68	1.36
	3D-Speck(0.25bpppb)	–	–	6.19	5.84
	EMPR (0.25bpppb)	–	–	61.23	69.15
	Mean-scale	17.60	155	0.22	0.45
	Context	20.42	158	3.28	9.01
	Cheng2020att	27.80	278	2.55	8.32
	CCM	24.77	259	0.34	8.6
	HMA	40.85	339	0.50	1.05
	CompNet (Base)	30.62	290	0.21	0.42
	HCCNet	41.37	495	0.2	0.41
HSImix	KLT+JPEG2000	–	–	1.21	0.76
	3D-Speck(0.25bpppb)	–	–	3.12	3.98
	EMPR(0.25bpppb)	–	–	31.91	29.88
	Mean-scale	17.78	47	0.06	0.10
	Context	20.60	48	0.44	1.43
	Cheng2020att	27.97	74	0.40	1.50
	CCM	24.94	68	0.1	2.2
	HMA	41.05	121	0.28	0.57
	CompNet (Base)	30.80	85	0.05	0.09
	HCCNet	41.74	146	0.06	0.11

representation. In addition, the model does not need PretextNet during the inference process; thus, the capacity will be the same as the baseline.

F. Discussion

For easy understanding, we make a discussion involving the following four points.

1) The proposed HCCNet model approaches the task of hyperspectral compression from the contrastive learning perspective to lessen the informative attribute collapses and losses with the help of CIFE and CIFR.
2) Figs. 8 and 9 illustrate that compression models can generate a good representation that benefits the compression performance via contrastive learning. The designed CIFE enables the latent representation to extract discriminative attributes, and the designed CIFR can help recover the lost attributes during decoding.
3) As shown in Figs. 10–19, both quantitative (RD performances) and visual comparisons demonstrate that the proposed HCCNet performs better than the existing compression methods in terms of the PSNR, MS-SSIM, and SAM. This advancement depends on the ability to capture discriminative attributes of the latent representation and relieve attribute loss. The classification results in Fig. 20 and Tables II and III demonstrate this point again.
4) HCCNet can be considered a self-supervised compression architecture with the aid of PretextNet. First, we can disentangle collapsed factors to obtain a discriminative latent representation via CIFE. In addition, the as-designed CIFR is useful in finding the lost attributes to compensate for information loss caused by rate–distortion optimization.

V. CONCLUSION

This article proposes an HCCNet to tackle attribute collapse and loss problems that occur in the latent representation caused by the inherent property of HSIs and quantization in compression. Instead of relying solely on the rate–distortion optimization to guide the learning of latent representation, we introduce positive and negative sample pairs into rate–distortion optimization to perform the optimization under a self-supervision architecture. Specifically, we design a CIFE module to keep the irrelevant features invariant and separate the relevant features to relieve attribute collapses. In addition, we define a CIFR module to guide the decoder in recovering the lost attributes by maximizing the mutual information between the original HSI and the reconstructed one from the feature level. Experiments on five different HSI datasets demonstrated that the proposed HCCNet could achieve superior compression performance from both quantitative and visual perspectives. However, the semantics of the latent representation that the HCCNet learned may not have been sufficiently relevant to the downstream task, which led to a semantic gap between the compressed HSI and the original one. Moving forward, we will explore some task-level features in latent representation learning to make the HSI compression method more appropriate for real-world requirements.

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